Tianlin Liu, Ph.D.

Texas Tech University • Department of Chemistry and Biochemistry

EDUCATION

Ph.D. in Chemistry

University of Pennsylvania, Philadelphia, PA — Sep. 2018–Nov. 2023

Thesis: *Electronic Spectroscopy, Photochemistry, and Unimolecular Reactions of Criegee Intermediates*
Advisor: Prof. Marsha I. Lester

B.S. in Chemistry

University of Science and Technology of China, Hefei, Anhui, China — Sep. 2014–Jun. 2018

Thesis: *Raman Spectroscopy of Graphene Electrode*
Advisor: Prof. Qun Zhang

PROFESSIONAL EXPERIENCE

Texas Tech University, Lubbock, TX — Starting Aug. 2026

Assistant Professor in the Chemistry and Biochemistry department

University of California, San Diego, La Jolla, CA — Nov. 2023–Jun. 2026

Postdoctoral researcher with Prof. Wei Xiong.

University of Pennsylvania, Philadelphia, PA — Jan. 2019–Nov. 2023

Graduate student research assistant with Prof. Marsha Lester.

University of Science and Technology of China, Hefei, Anhui, China — Sep. 2017–Jun. 2018

Undergraduate research assistant with Prof. Qun Zhang.

University of Wisconsin, Madison, Madison, WI — Jun. 2017–Aug. 2017

Undergraduate research assistant through the REU program with Prof. Gilbert Nathanson.

University of Science and Technology of China, Hefei, Anhui, China — Sep. 2016–Jun. 2017

Undergraduate research assistant with Prof. Zhaoxiang Deng.

AWARDS

ACS Physical Chemistry Young Investigator Award — 2025
Silver medal in 27th Chinese Chemical Olympiad (CChO) — 2013

PUBLICATIONS (# denotes equal contribution. * denotes corresponding author(s))

- Yin, G.#; **Liu, T.#***; Zhang, L., Sheng, T., Mao, H. and Xiong, W.*, Overcoming energy disorder for cavity-enabled energy transfer in vibrational polaritons, *Science*, 2025, 389, 845-848.
- **Liu, T.**; Yin, G. and Xiong, W.*, Unlocking Delocalization: How Much Coupling Strength can Overcome Energy Disorder in Molecular Polaritons? *Chem. Sci.*, 2025, 16, 4676-4683.
- **Liu, T.**; Elliott, S.N.; Zou, M.; Vansco, M.F.; Sojdak, C.A.; Markus, C.R.; Almeida, R.; Au, K.; Sheps, L.; Osborn, D.L.; Winiberg, F.A.F.; Percival, C.J.; Taatjes, C.A.; Caravan, R.C.; Klippenstein, S.J.* and Lester, M.I.,* OH roaming and beyond in the unimolecular decay of the methyl-ethyl substituted Criegee intermediate: observations and predictions, *J. Am. Chem. Soc.*, 2023, 145, 35, 19405-10420.
- **Liu, T.#**; Roy, T.K.#; Qian, Y.; Sojdak, C.A.; Kozlowski, M.C. and Lester, M.I.*, A five-carbon unsaturated Criegee intermediate: synthesis, spectroscopic identification, and theoretical study of 3-penten-2-one oxide, *Chem. Sci*., 2023, 14, 10471-10477.

- **Liu, T.** and Lester, M.I.*, Roaming in the unimolecular decay of *syn*-methyl substituted Criegee intermediates, *J. Phys. Chem. A*, 2023, 127, 51, 10817-10827.
- Wang, G.#; **Liu, T.#**; Zou, M.; Karsili, T.N.V. and Lester, M.I.*, UV photodissociation dynamics of the acetone oxide Criegee intermediate: experiment and theory, *Phys. Chem. Chem. Phys.,* 2023, 25, 7453-7465.
- Wang, G.#; **Liu, T.#**; Zou, M.; Sojdak, C.A.; Kozlowski, M.C.; Karsili, T.N.V. and Lester, M.I.*, Electronic spectroscopy and dissociation dynamics of vinyl-substituted Criegee intermediates: 2-butenal oxide and comparison with methyl vinyl ketone oxide and methacrolein oxide isomers, *J. Phys. Chem. A*, 2023, 127, 1, 203–215.
- **Liu, T.#**; Zou, M.#; Caracciolo, A.; Sojdak, C.A. and Lester, M.I.*, Substituent effects on the electronic spectroscopy of four-carbon Criegee intermediates *J. Phys. Chem. A*, 2022, 126, 38, 6734-6741.
- Karlsson, E.; Rabayah, R.; **Liu, T.**; Cruz, E.M.; Kozlowski, M.C; Karsili, T.N.V. and Lester, M.I.*, Electronic spectroscopy of the halogenated Criegee intermediate, ClCHOO: Experiment and theory, *J. Phys. Chem. A*, 2024, 128, 51, 10949-10956.
- Zou, M.; **Liu, T.**; Vansco, M.F.; Elliott, S.N.; Sojdak, C.A.; Markus, C.R.; Almeida, R.; Au, K.; Sheps, L.; Osborn, D.L.; Winiberg, F.A.F.; Percival, C.J.; Taatjes, C.A.; Klippenstein, S.J.; Lester, M.I. and Caravan, R.L.*, Bimolecular reaction of methy-ethyl substituted Criegee intermediate with SO_2, *J. Phys. Chem. A*, 2023, 127, 43, 8994-9002.
- Wang, G.; **Liu, T.**; Caracciolo, A.; Vansco, M.F.; Trongsiriwat, N.; Walsh, P.J.; Marchetti, B.; Karsili, T.N.V. and Lester, M.I.*, Photodissociation dynamics of methyl vinyl ketone oxide: a four-carbon unsaturated Criegee intermediate from isoprene ozonolysis. *J. Chem. Phys*., 2021, 155, 174305.
- Esposito, V.J.; **Liu, T**.**#**; Wang, G.#; Caracciolo, A.; Vansco, M.F.; Marchetti, B.; Karsili, T.N.V. and Lester, M.I.*, Photodissociation dynamics of CH_2OO on multiple potential energy surfaces: experiment and theory. *J. Phys. Chem. A*, 2021, 125, 30, 6571-6579.

PRESENTATIONS

- "Ultrafast Vibrational Spectroscopy of Nanoconfined Water in Beryl", Oral presentation, Vibrational Spectroscopy Gordon Research Seminar, New Hampshire, 2026.
- "Overcoming disorder to unlock energy transfer in vibrational polaritons", Oral presentation, ACS Fall meeting, Washington DC, 2025.
- "Understanding polariton chemistry from an ultrafast perspective", Oral presentation, ACS Spring meeting, San Diego, CA, 2024.
- "H-bond dynamics modulated by vibrational polaritons: Inhomogeneous broadening matters", Poster presentation, 2025 ACS Spring meeting, San Diego, CA, 2024.
- "Energy disorder and delocalization of vibrational polaritons", Poster presentation, SPIE Optics + Photonics, San Diego, CA, 2024.
- "Complexation and Vibrational Energy Transfer of the 2,6-ditertbutylphenol under Vibrational Strong Coupling Conditions", Poster presentation, Multidisciplinary University Research Initiative (MURI) Program Review, Arlington, VA, 2024.
- "OH roaming in the unimolecular decay of the alkyl-substituted Criegee intermediates", Poster presentation, 28^{th} Dynamics of Molecular Collisions Conference, Snowbird, UT, 2023.
- "Novel OH roaming pathway in the unimolecular decay of alkyl-substituted Criegee intermediates", Poster presentation, Gordon Research Conference on Molecular Interactions and Dynamics, Easton, MA, 2022.
- "Substituent effects on the electronic spectroscopy of four-carbon Criegee intermediates", Poster presentation, American Geophysical Union Fall Meeting, 2021.

RESEARCH EXPERIENCE

Postdoctoral Researcher, University of California San Diego

Combined experimental and computational studies on vibrational strong coupling.

- Experimentally investigated the ultrafast dynamics and nonlinear properties of vibrational polaritons using two-dimensional infrared spectroscopy (2D IR) for novel chemical and photonic applications, such as polariton-enabled energy transfer in H-bonded phenol systems.

- Theoretically investigated the delocalization property of vibrational polaritons in the presence of energy disorder and proposed new delocalization criteria for vibrational strong coupling.

Graduate Research Assistant, University of Pennsylvania

Combined experimental and computational studies of the photochemistry, unimolecular and bimolecular reactivity on atmospherically relevant reactive intermediates from alkene ozonolysis (Criegee intermediate).

- Investigated the electronic transition and excited state chemistry of the jet-cooled Criegee intermediates in vacuum using mass spectrometry and velocity map imaging (VMI), and performed advanced quantum chemistry calculation.
- Identified a novel unimolecular OH-roaming pathway for Criegee intermediate using the Sandia multiplexed photoionization mass spectrometer apparatus through collaboration with Lawrence-Berkeley National Laboratory (LBNL) Advanced Light Source (ALS).

MENTORSHIP AND TEACHING EXPERIENCE

Postdoctoral Research Assistant, University of California San Diego

- Mentored two graduate students on experimental, theoretical work, and scientific writing.

Graduate Research Assistant, University of Pennsylvania

- Provided training on laboratory skills to two postdoctoral researchers and mentored two graduate students by supervising experimental work and instructing quantum chemistry calculations.

Graduate Teaching Assistant, University of Pennsylvania

- Courses including General Chemistry Lab, General Chemistry, Physical Chemistry 1 – Quantum Mechanics, and Environmental Chemistry. Duty including supervising laboratory sessions and giving recitation classes.

VOLUNTEER WORK

- Volunteered for Graduate student recruitment Open House Weekend at the University of Pennsylvania in 2022 and 2023.
- Volunteered for Graduate student recruitment Open House Weekend at the University of California San Diego in 2025.
- Volunteered as the Macao team coordinator during the 29th Chinese Chemical Olympiad (Finals) in 2015.